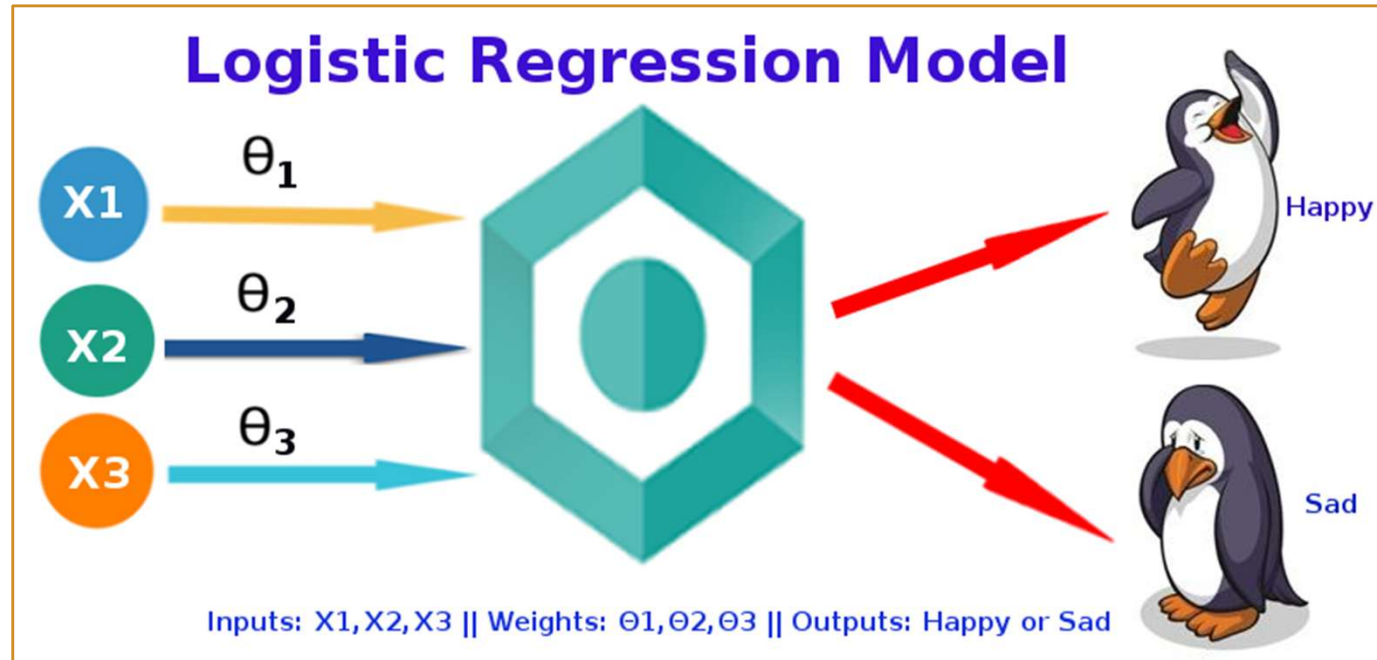




Logistic Regression

Introduction

Logistic regression is a fundamental supervised machine learning algorithm used for binary classification. Instead of predicting continuous numerical values like linear regression, it calculates the probability that a given data point belongs to a specific category (e.g., true/false or yes/no)



Working

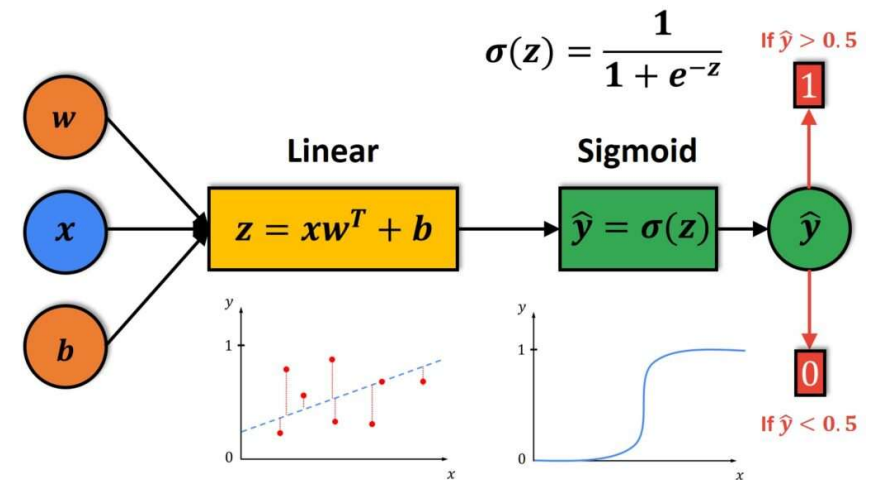
How it Works

❖ Instead of fitting a straight line to data, logistic regression applies the mathematical logistic (sigmoid) function to map predictions to an S-shaped curve

- **The Output:** It yields a probability score ranging between (0) and (1).
- **The Decision Boundary:** A threshold (usually (0.5)) is set. Probabilities above the threshold map to the positive class ((1)); those below map to the negative class ((0)).

➤ **The Math:**

$$P = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}}$$



Common Use Cases

Logistic regression is widely used across various industries due to its interpretability and speed:

- ❖ **Medicine:** Predicting the presence or absence of a disease based on patient vitals and test results.
- ❖ **Finance:** Assessing the likelihood of a homeowner defaulting on a mortgage.
- ❖ **Marketing/Tech:** Classifying whether an incoming email is spam or not spam.

Applications of Binary Logistic Regression

Customer Churn Analysis

Customer churn analysis is simple yet crucial for business strategy.



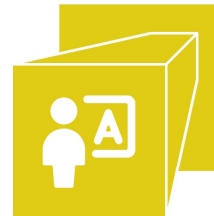
Disease Prediction

Disease prediction requires complex analysis for high-impact outcomes.



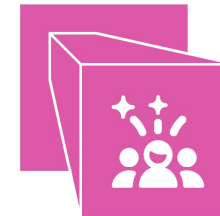
Basic Classification Tasks

Basic classification tasks are straightforward with minimal impact.



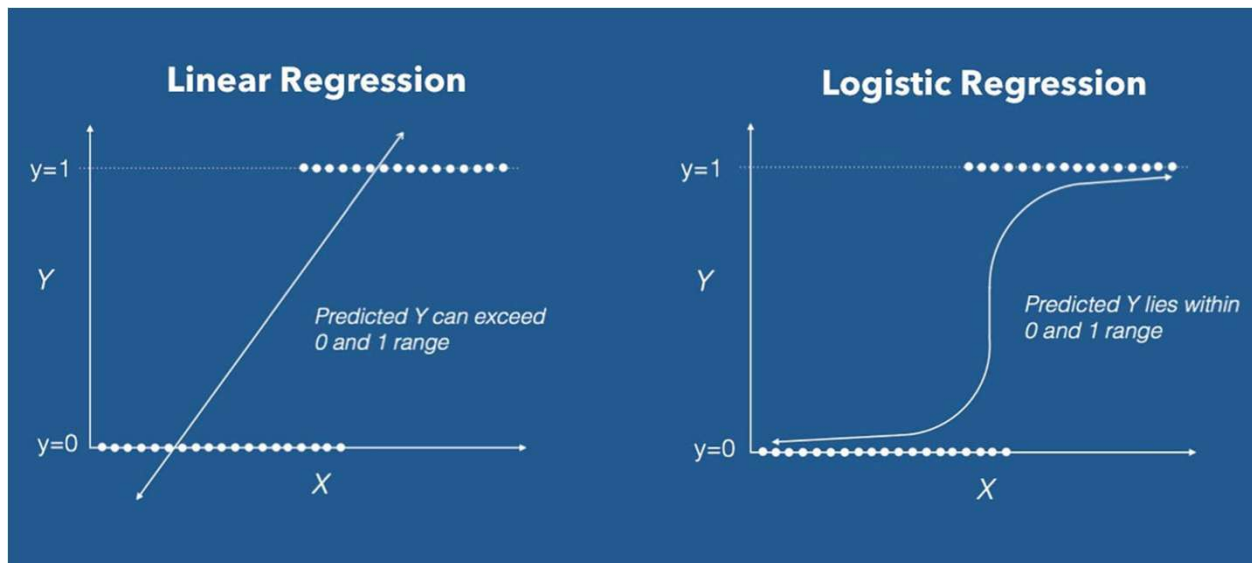
Advanced Marketing Strategies

Advanced marketing strategies involve complex methods with limited impact.



Key Differences from Linear Regression

Feature 	Linear Regression	Logistic Regression
Target Variable	Continuous (e.g., salary, price)	Categorical (e.g., pass/fail, spam/ham)
Prediction Range	$-\infty$ to $+\infty$	0 to 1 (Probability)
Fitting Method	Ordinary Least Squares (OLS)	Maximum Likelihood Estimation (MLE)



Thank you!

